STA2001 Probability and Statistics I

Assignment 5

Deadline: 23:59 pm, Oct 17th, Friday

1. Consider a hypergeometric distribution with parameters (N, N_1, n) and a binomial distribution with parameters $(n, p = N_1/N)$. Use your preferred software to compute and compare the probability mass functions (pmfs) of these two distributions for the following parameters. In particular, what can you say about the differences between these two distributions as N gets larger?

(a) $n = 10$, $N_1 = 6$ and $N = 20$.
(b) $n = 10$, $N_1 = 30$ and $N = 100$.
(c) $n = 10$, $N_1 = 300$ and $N = 1000$.

几何分布模型.

2. In 2012, Red Rose tea randomly began placing 1 of 12 nautical figurines in a 100-bag box of the tea.

(a) On the average, how many boxes of tea must be purchased by a customer to obtain a complete collection consisting of the 12 nautical figurines? Hint: Let X be the number of boxes must be purchased by a customer to obtain a complete collection, and X_k be the number of boxes we need to purchase in order to have a nautical figurine that is different from the k nautical figurines we already got. Then $X = 1 + X_1 + \cdots + X_{11}$.

(b) If the customer uses one tea bag per day, how long can a customer expect to take, on the average, to obtain a complete collection?

每个X_i都是几何分布!!

3. Suppose the probability of suffering a fever from the flu vaccine is 0.005. If 1000 people are given the vaccine, use the Poisson distribution to approximate the probability that (a) 1 person suffers a fever as a result; and (b) more than 6 people suffer a fever as a result.

(Possion(5))

4day: Possion $\mu=5$ $\quad f_X(x)=e^{-5}\cdot\frac{5^x}{x!}$

4. A factory uses tools of a particular type. From time to time failures in these tools occur and they need to be replaced. The number of such failures in a day has a Poisson distribution with mean 1.25. At the beginning of a particular day there are five replacement tools in stock. A new delivery of replacements will arrive after four days. If all five spares are used before the new delivery arrives then further replacements cannot be made until the delivery arrives. Find

(a) the probability that three replacements are required over the next four days;

(b) the expected number of replacements actually made over the next four days.

(a.) $f_X(3)=\sim$

(b.) 需求 ~ Possion; 消耗 ≤5时 = 需求, ≥6后 fixed. $\Rightarrow E=5+$

5. An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil.

(a) What is the probability that the third strike comes on the seventh well drilled?

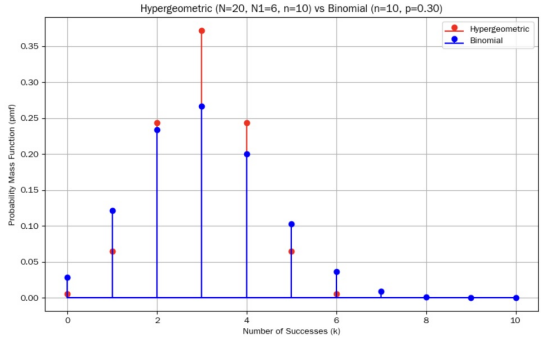
(b) What is the mean and variance of the number of wells that must be drilled if the oil company wants to set up three producing wells?

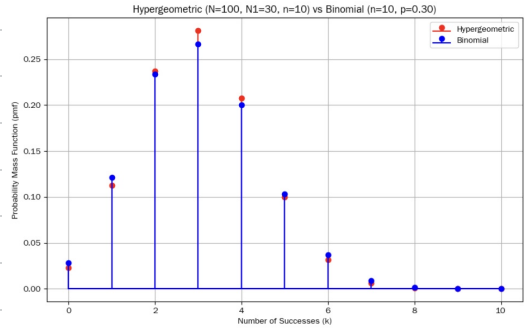
Ass5. Li Quan 124090310.

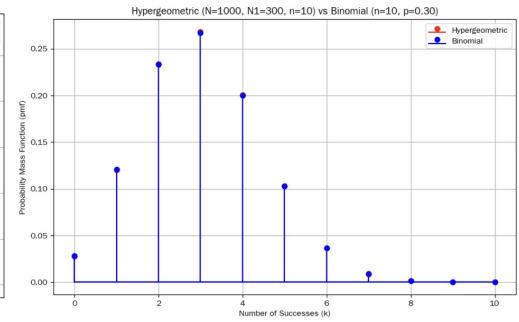
T1. I use numpy to do a visual graph. (the python code is attached in my zip.)

When N gets larger, the difference generated by replacement will be very small. So, hypergeometric distribution will converge to the binomial distribution.

- For N=20, N1=6, n=10, Sum of absolute differences in pmf: 0.3068
- For N=100, N1=30, n=10, Sum of absolute differences in pmf: 0.0511
- For N=1000, N1=300, n=10, Sum of absolute differences in pmf: 0.0049







T2. (a.) X_k satisfies geometric distribution. $p_k=\frac{12-k}{12}$, $E(X_k)=\frac{1}{p_k}=\frac{12}{12-k}$

$\Rightarrow X=1+X_1+X_2+\cdots+X_{11}$, $E(X)=1+E(X_1)+\cdots+E(X_{11})=1+\frac{12}{11}+\frac{12}{10}+\cdots+\frac{12}{1}\approx 37.2385$ $\Rightarrow$ buy about 38 boxes of tea on average.

(b.) $37.2385\times 100=3723.85\approx 3724$ $\Rightarrow$ need 3724 days on average.

T3. (a.) $\lambda=np=1000\times 0.005=5$ $P(X=1)=\frac{e^{-5}\cdot 5^1}{1!}\approx 0.0337$

(b.) $P(X>6)=1-P(X\le 6)=1-\sum_{k=0}^{6}\frac{e^{-5}\cdot 5^k}{k!}=1-0.76218346\approx 0.2378$

T4. (a.) 1 day $\rightarrow$ poisson with $\lambda=1.25$, four days $\rightarrow$ poisson with $\lambda=5$. X: number of required replacements. Y: number of actual replacements.

$P(X=3)=\frac{e^{-5}\cdot 5^3}{3!}=\frac{125\cdot e^{-5}}{6}\approx 0.1404$

(b.) $Y=X$ if $X\le 5$, $Y=5$ if $X>5$.

$E(Y)=\sum_{k=0}^{5}\frac{e^{-5}\cdot 5^k}{k!}\cdot k+\sum_{k=6}^{\infty}\frac{e^{-5}\cdot 5^k}{k!}\cdot 5\approx 4.12$

T5. (a.) negative binomial distribution. $P(X=7)=C_6^2\cdot(\frac{1}{5})^2\cdot(\frac{4}{5})^4\cdot\frac{1}{5}=\frac{6!}{2!\times 4!}\times 0.008\times 0.4096=0.049152$

(b.) $r=3$. $p=0.2$ $\Rightarrow E(X)=\frac{r}{p}=3\times 5=15$.

$Var(X)=\frac{r(1-p)}{p^2}=\frac{3\times(1-0.2)}{(0.2)^2}=60$